

不定积分思考题

$$I = \int \frac{2 \cdot 3^x - 5 \cdot 2^x}{3^x} dx = 2x - \frac{5(2/3)^x}{\ln(2/3)} + C;$$

$$1、 I = \int \sec x (\sec x - \tan x) dx = \tan x - \sec x + C;$$

$$2、 I = \int \cos^2 \frac{x}{2} dx = \frac{1}{2}(x + \sin x) + C;$$

$$3、 I = \int \frac{dx}{1 + \cos 2x} = \frac{1}{2} \int \frac{dx}{\cos^2 x} = \frac{1}{2} \tan x + C;$$

$$4、 I = \int \frac{\cos 2x}{\cos x - \sin x} dx = \sin x - \cos x + C;$$

$$4、 I = \int \frac{\sin \sqrt{t}}{\sqrt{t}} dt = -2 \cos \sqrt{t} + C;$$

$$5、 I = \int \tan^{10} x \sec^2 x dx = \frac{1}{11} \tan^{11} x + C;$$

$$6、 I = \int \frac{dx}{x \ln x \ln \ln x} = \ln \ln \ln x + C;$$

$$7、 I = \int \frac{x}{\sqrt{1+x^2}} \tan \sqrt{1+x^2} dx = -\ln |\cos \sqrt{1+x^2}| dx + C; (*)$$

$$8、 I = \int \frac{dx}{\sin x \cos x} = \ln |\csc 2x - \cot 2x| + C;$$

$$9、 I = \int \frac{dx}{e^x + e^{-x}} = \arctan e^x + C; (*)$$

$$10、 I = \int x e^{-x^2} dx = -\frac{1}{2} e^{-x^2} + C;$$

$$11、 I = \int x \cos x^2 dx = \frac{1}{2} \sin x^2 + C;$$

$$12、 I = \int \frac{x}{\sqrt{2-3x^2}} dx = -\frac{1}{3} \sqrt{2-3x^2} + C;$$

$$13、 I = \int \frac{\sin x + \cos x}{\sqrt[3]{\sin x - \cos x}} dx = -\frac{3}{2} (\sin x - \cos x)^{2/3} + C;$$

$$14、 I = \int \frac{1-x}{\sqrt{9-4x^2}} dx = \frac{1}{2} \arcsin \left(\frac{2}{3} x \right) + \frac{1}{4} \sqrt{9-4x^2} + C;$$

$$15、 I = \int \frac{x^3}{9+x^2} dx = \frac{1}{2} (x^2 - 9 \ln(9+x^2)) + C;$$

$$16、 I = \int \frac{dx}{2x^2 - 1} = \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2}x - 1}{\sqrt{2}x + 1} \right| + C ;$$

$$17、 I = \int \cos^3 x dx = \sin x - \frac{1}{3} \sin^3 x + C ;$$

$$18、 I = \int \sin 2x \cos 3x dx = \frac{1}{2} \cos x - \frac{1}{10} \cos 5x + C ;$$

$$19、 I = \int \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x} ;$$

$$20、 I = \int e^{\sin^2 x} \sin 2x dx = e^{\sin^2 x} + C ;$$

$$21、 I = \int \frac{1 - \tan x}{1 + \tan x} dx = \ln |\cos x + \sin x| + C$$

$$22、 I = \int \frac{dx}{\sin 2x + 2 \sin x}$$

$$I = \frac{1}{4} \int \left(\frac{1}{t} + t \right) dt = \frac{1}{4} \ln \left| \tan \frac{x}{2} \right| + \frac{1}{8} \tan^2 \frac{x}{2} + C$$

$$25、 I = \int \sqrt{\frac{e^x - 1}{e^x + 1}} dx$$

$$I = \ln \left| e^x + \sqrt{e^{2x} - 1} \right| + \arcsin e^{-x} + C$$

$$26、 I = \int \frac{dx}{\sqrt{x(4-x)}} = -\arcsin \frac{2-x}{2} + C$$

$$27、 I = \int \frac{1}{x} \sqrt{\frac{x}{4-x}} dx ;$$

$$I = 2 \arctan \sqrt{\frac{x}{4-x}} + C$$

$$28、 I = \int \frac{\ln \tan x}{\sin 2x} dx$$

$$I = \frac{1}{4} \ln^2 \tan x + C$$

$$29、 I = \int \sqrt{e^{2x} + 4e^x - 1} dx ; (*)$$

$$I = \sqrt{e^{2x} + 4e^x - 1} + 2 \ln \left| e^x + 2 + \sqrt{e^{2x} + 4e^x - 1} \right| - \arcsin \frac{2 - e^{-x}}{\sqrt{5}} + C$$

$$30、 I = \int \frac{10^{2 \arccos x}}{\sqrt{1-x^2}} dx = -\frac{10^{2 \arccos x}}{2 \ln 10} + C$$

$$31、 I = \int \tan^3 x \sec x dx = \frac{1}{3} \sec^3 x - \sec x + C$$

$$32、 I = \int \frac{\arctan x dx}{1+x^2} = \frac{1}{2} (\arctan x)^2 + C$$

$$33、 I = \int \frac{\arctan \sqrt{x}}{\sqrt{x}(1+x)} dx = 2 \int \arctan \sqrt{x} d(\arctan \sqrt{x}) + C$$

$$34、 I = \int \frac{\arctan \sqrt{x}}{\sqrt{1+x}} dx$$

$$I = 2\sqrt{x+1} \arctan \sqrt{x} - \ln|\sqrt{x+1} + \sqrt{x}| + C$$

$$35、 I = \int \frac{dx}{(\arcsin x)^2 \sqrt{1-x^2}}$$

$$36、 I = \int \frac{x^2}{\sqrt{a^2-x^2}} dx ;$$

$$I = \frac{a^2}{2} \left(\arcsin \frac{x}{a} - \frac{1}{a^2} x \sqrt{a^2-x^2} \right) + C$$

$$37、 I = \int \frac{dx}{\sqrt{(x^2+1)^3}} ;$$

$$I = \frac{x}{\sqrt{x^2+1}} + C$$

$$38、 I = \int \frac{dx}{1+\sqrt{2x}} = \sqrt{2x} - \ln(1+\sqrt{2x}) + C ; \quad (\text{令 } t = \sqrt{2x})$$

$$39、 I = \int \frac{dx}{1+\sqrt{1-x^2}} = \arcsin x - \frac{1-\sqrt{1-x^2}}{x} + C$$

$$40、 I = \int \frac{dx}{x+\sqrt{1-x^2}}$$

$$I = \frac{1}{2} \left(\arcsin x + \ln|x + \sqrt{1-x^2}| \right) + C$$

$$41、 I = \int \ln \left(1 + \sqrt{\frac{1+x}{x}} \right) dx$$

$$I = \frac{1}{4} \ln \left| \sqrt{\frac{1+x}{x}} - 1 \right| - \frac{1}{4} \ln \left| \sqrt{\frac{1+x}{x}} + 1 \right| + \frac{\sqrt{x}}{2(\sqrt{x} + \sqrt{1+x})} + C$$

$$42、 I = \int \frac{\ln(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx = \frac{1}{2} \ln^2(x + \sqrt{1+x^2}) + C$$

$$43、 I = \int \frac{\ln x}{x\sqrt{1+\ln x}} dx$$

$$44、 I = \int \frac{\sqrt{\ln(1+x) - \ln x}}{x(x+1)} dx ;$$

$$45、 I = \int \frac{\sin^2 x - \cos^2 x}{\sin^4 x \cos^4 x} dx ;$$

$$I = -\frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2} + \sin 2x}{\sqrt{2} - \sin 2x} \right| + C$$

$$46、 I = \int \frac{e^{\sin 2x} \sin^2 x}{e^{2x}} dx ;$$

$$I = -\frac{1}{4} e^{\sin 2x - 2x} + C$$

$$47、 I = \int \frac{x \ln(1 + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx$$

$$I = (1 + \sqrt{1+x^2}) \ln(1 + \sqrt{1+x^2}) - \sqrt{1+x^2} + C$$

$$48、 I = \int e^{2x} (\tan x + 1)^2 dx$$

$$49、 I = \int \frac{1 - \ln x}{(x - \ln x)^2} dx$$

$$I = \frac{x}{x - \ln x} + C$$

$$50、 I = \int \frac{x e^x}{\sqrt{e^x - 1}} dx$$

$$I = 2x\sqrt{e^x - 1} - 4\sqrt{e^x - 1} + 4 \arctan \sqrt{e^x - 1} + C$$

$$51、 f(x) = \begin{cases} x+1, & x \leq 0 \\ e^{2x}, & x > 0 \end{cases}, \text{ 求 } I = \int f(x) dx$$

$$I = \begin{cases} x^2/2 + x + 1/2 + C, & x \leq 0 \\ e^{2x}/2 + C, & x > 0 \end{cases}$$

$$52、 I = \int \frac{dx}{2 \sin x + 3 \cos x};$$

$$I = \frac{1}{\sqrt{13}} \int \frac{dx}{\cos(x+\varphi)} = \frac{1}{\sqrt{13}} \ln |\sec(x+\varphi) + \tan(x+\varphi)| + C \quad (\varphi = \arctan \frac{3}{2})$$

$$53、 I = \int \frac{x + \sin x}{1 + \cos x} dx$$

$$I = x \tan \frac{x}{2} + C$$

$$54、 \text{设 } F(x) \text{ 是 } f(x) \text{ 的一个原函数, } F(x) > 0, F(0) = 1$$

$$\textcircled{1} \text{ 设 } f(x) = \frac{x F(x)}{1+x^2}, \text{ 求 } f(x); \textcircled{2} f(x) F(x) = e^x, \text{ 求 } f(x)$$

$$55、 I = \int \sqrt{\frac{4-x}{4+x}} dx$$

$$I = \sqrt{16-x^2} - 8 \arctan \sqrt{\frac{4-x}{4+x}} + C$$

$$56、 I = \int \frac{x^2 \arctan x}{1+x^2} dx$$

$$I = x \arctan x - \frac{1}{2} \ln(1+x^2) - \frac{1}{2} \arctan^2 x + C$$

$$57、 I = \int \frac{x \arctan x}{(1+x^2)^2} dx$$

$$I = -\frac{\arctan x}{2(1+x^2)} + \frac{x}{4(1+x^2)} + \frac{1}{4} \arctan x + C$$

$$58、 I = \int x^2 \sqrt{\frac{a-x}{a+x}} dx$$

$$I = \frac{a^3}{2} \arcsin \frac{x}{a} - \frac{a}{2} x \sqrt{a^2 - x^2} - \frac{1}{3} (a^2 - x^2)^{3/2} + a^2 \sqrt{a^2 - x^2} + C$$

$$59、 I = \int \frac{1}{\sin(x+a) \sin(x+b)} dx, (a \neq b)$$

$$I = \frac{1}{\sin(a-b)} \ln \left| \frac{\sin(x+b)}{\sin(x+a)} \right| + C$$

$$60、 I = \int \frac{dx}{x\sqrt{x^2-1}};$$

$$I = \arccos \frac{1}{x} + C$$

$$61、 I = \int \frac{dx}{x(x^n+1)}$$

$$I = \frac{1}{n} \ln \left| \frac{x^n}{x^n+1} \right| + C;$$

$$62、 I = \int \left[\frac{f(x)}{f'(x)} - \frac{f''(x)f^2(x)}{f'^3(x)} \right] dx$$

$$63、 I = \int \frac{x+1}{x(1+xe^x)} dx;$$

$$I = \ln \left| \frac{xe^x}{1+xe^2} \right| + C.$$

$$64、 I = \int \frac{\ln^2 x}{x^3} dx$$

$$I = -\frac{\ln^2 x}{2x^2} - \frac{\ln x}{2x^2} - \frac{1}{4x^2} + C$$

$$65、 I = \int x^3 f'(x) dx, \text{ 其中 } \int f(x) dx = \frac{\sin x}{x};$$

$$66、 \text{ 设 } f(\ln x) = \frac{\ln(1+x)}{x}, \text{ 求 } I = \int f(x) dx$$

$$67、 I = \int \frac{x^2 e^{2x}}{(1+x)^2} dx;$$

$$68、 I = \int \frac{x}{x+\sqrt{x^2-1}} dx$$

$$I = \frac{1}{3} x^3 - \frac{1}{3} (x^2-1)^{3/2} + C$$

$$69、 I = \int \frac{\sqrt{\ln(x+\sqrt{1+x^2})+5}}{\sqrt{1+x^2}} dx;$$

$$70、 I = \int \frac{\arctan e^x}{e^x} dx;$$

71、 $I_n = \int \sec^n x dx$, 证明递推公式:

$$I_n = \frac{1}{n-1} \sec^{n-2} x \cdot \tan x + \frac{n-2}{n-1} I_{n-2} \quad (n \geq 2)$$

72、 $I = \int |x-1| dx$

$$I = \int |x-1| dx = \begin{cases} x - \frac{1}{2}x^2 + 1 + C, & x \leq 1 \\ \frac{1}{2}x^2 - x + C, & x > 1 \end{cases}$$

73、 $I = \int \frac{dx}{1 + e^{\frac{x}{2}} + e^{\frac{x}{3}} + e^{\frac{x}{6}}}$;

$$I = x - 3 \ln \left(e^{\frac{x}{6}} + 1 \right) - \frac{3}{2} \ln \left(e^{\frac{x}{3}} + 1 \right) - 3 \arctan e^{\frac{x}{6}} + C$$

74、 $I = \int \frac{3 \cos x - \sin x}{\cos x + \sin x} dx$

75、 $I_1 = \int \frac{\sin x}{a \cos x + b \sin x} dx$ 及 $I_2 = \int \frac{\cos x}{a \cos x + b \sin x} dx$

$$I_1 = \frac{1}{a^2 + b^2} (bx - a \ln |a \cos x + b \sin x|) + C ,$$

$$I_2 = \frac{1}{a^2 + b^2} (ax + b \ln |a \cos x + b \sin x|) + C ,$$